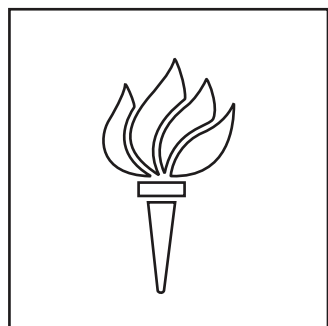


NYU Ability Project

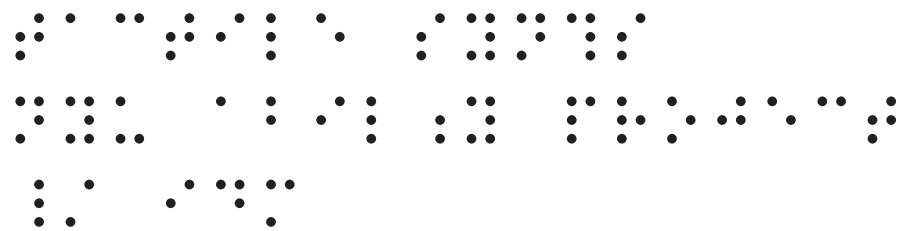
## Serge Modular 73-75 Panels



NYU

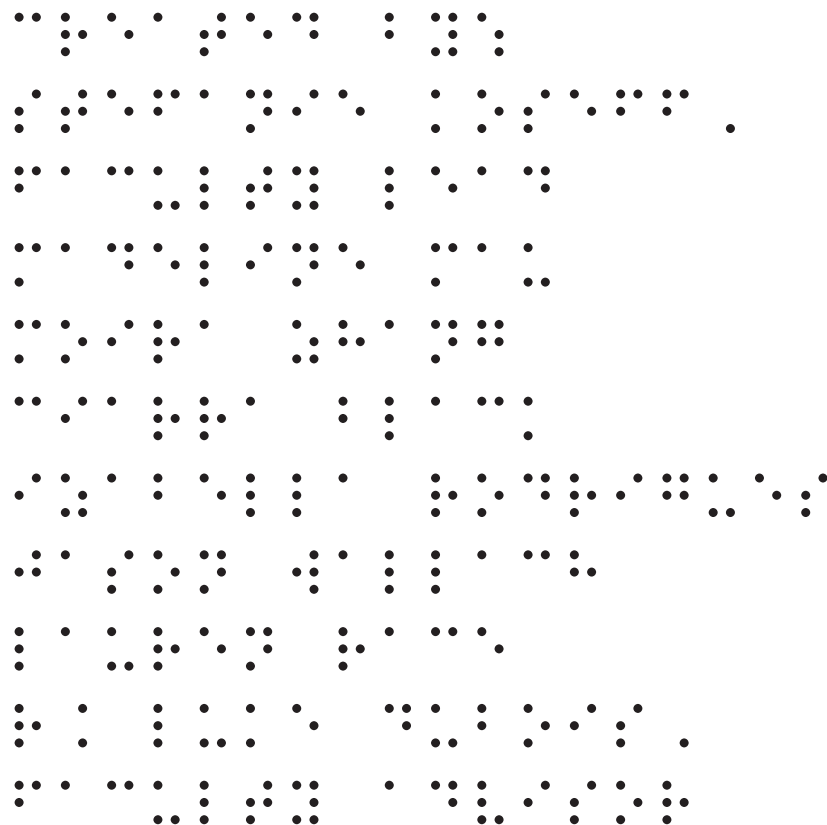
Ability  
Project

NYU Ability Project



Tactile Synths

NYU Ability Project / IDM



Created by:

Stefanie Koseff, Faculty Lead

Madeline Mau

Moirra Zhang

Ciarra Black

Izabella Rodrigues

Jason Wallach

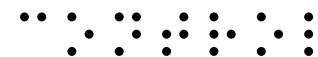
Lauren Race

R. Luke DuBois, Faculty Advisor

# panel guide



VOICE



CONTROL



HOMEBUILT #1



HOMEBUILT #2



PRESET



OSC



OSC



TWS



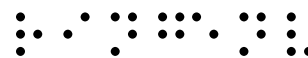
VCF



VCA



VCA



RING



ENV



ADP



NOI



POS



NEG



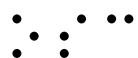
COM



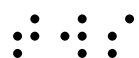
PRC



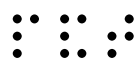
SSG



OSC



TWS



P&T



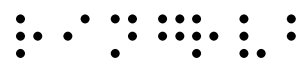
COM



PRC



VCA



RING



RVB



PRE



POS



NEG



ENV



ENV



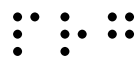
ENV



ADP



SEQ



PRG



TBS

# table of contents

|     |   |      |    |     |    |     |    |
|-----|---|------|----|-----|----|-----|----|
| KEY | 1 | PRC  | 6  | POS | 11 | VCF | 16 |
| OSC | 2 | VCA  | 7  | NEG | 12 | SSG | 17 |
| TWS | 3 | RING | 8  | ENV | 13 | SEQ | 18 |
| P&T | 4 | RVB  | 9  | ADP | 14 | PRG | 19 |
| COM | 5 | PRE  | 10 | NOI | 15 | TBS | 22 |

# KEY (1/2)

## symbol key



ac input



misc input



ac output



misc output



dc input



basic knob



dc output



offset knob



pulse input



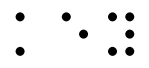
scale knob

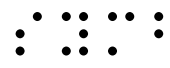


pulse output



button

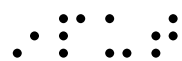



**KEY (2/2)**  



 symbol key



trs input



trs output



OSC

oscillator



sine ac



ws x



sine dc



waveshape



saw dc



saw ac



fm



fm



fm x



fm x



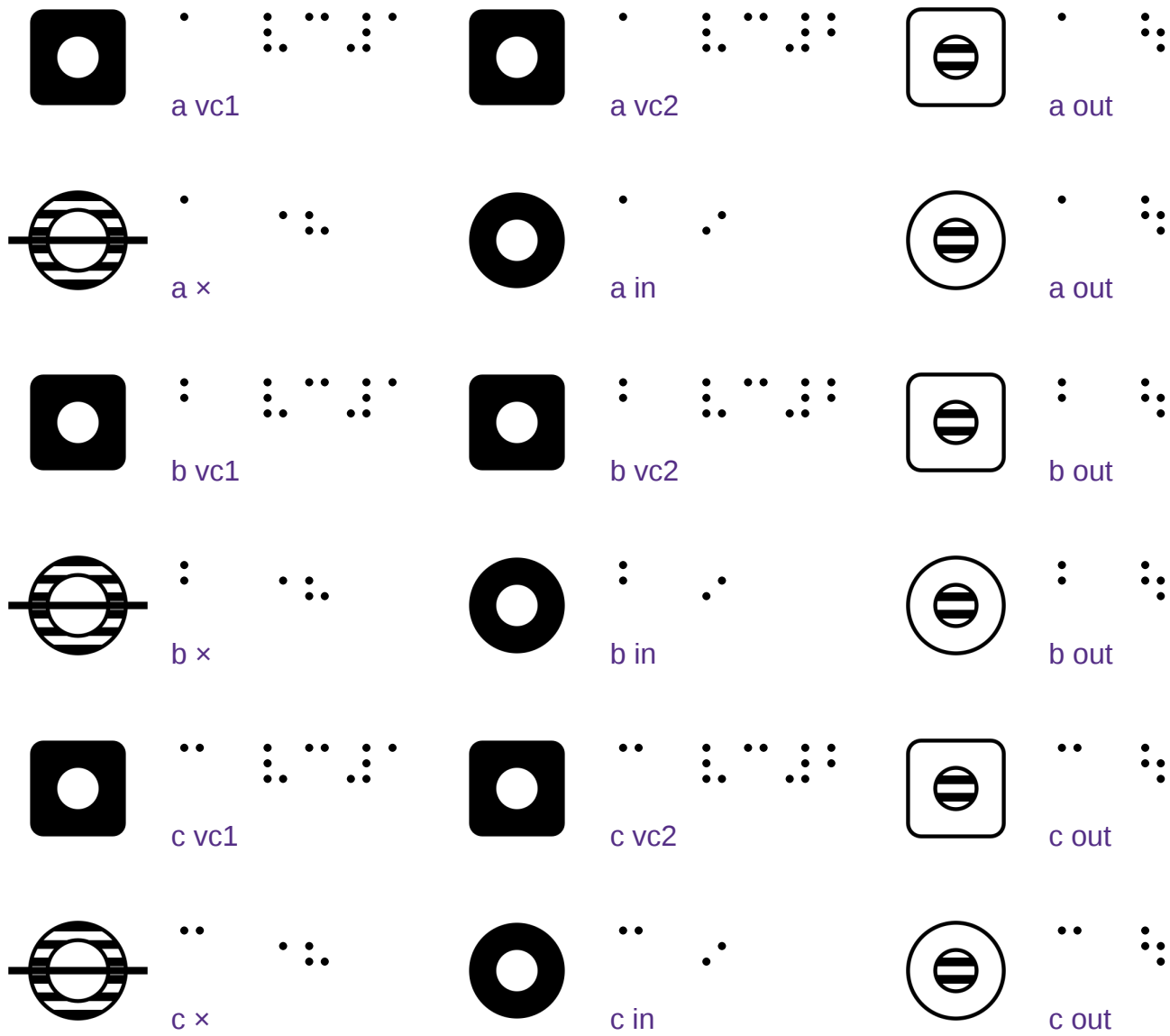
sync



frequency



triple waveshaper





P&T

## peak & trough



peak out



trough out



peak out



trough out



peak in



trough in



peak in



trough in



peak in



trough in



peak in



trough in

COM

triple comparator



a in 1



a out



a thresh



a in 2



b in 1



b out



b thresh



b in 2



c in 1



c out



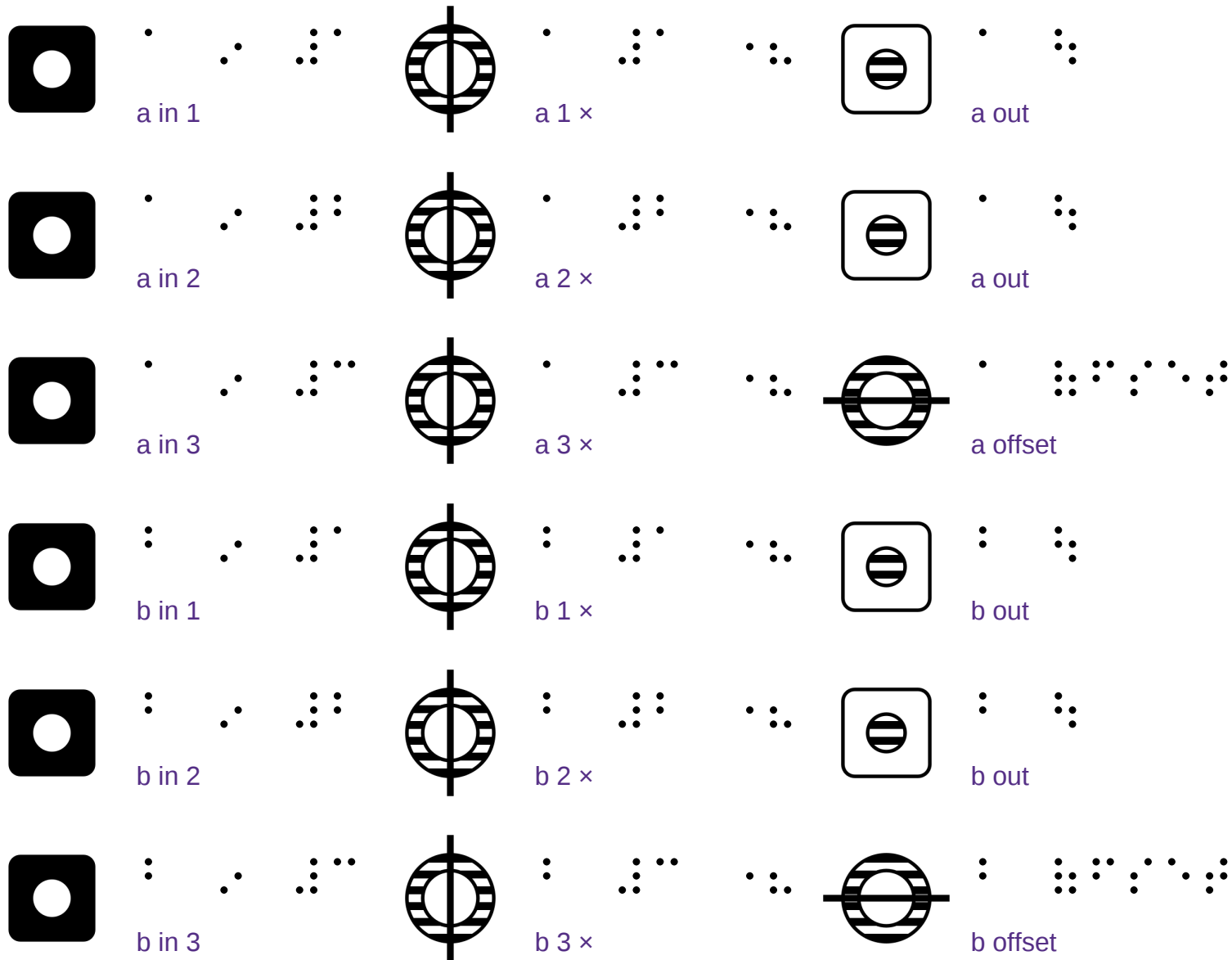
c thresh



c in 2

PRC

dual processor



# VCA gate



out



ac



dc



lin



log



gain

ring modulator



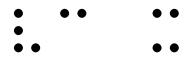
out

 $y$ 

vc y



X



VC X

 $x - xy$

RVB

reverbulator



out



out



mix



wet



wet



in



preamplifier



input



gain



out



out

POS

positive slew



a vc



a in



a out



a vc x



a start



a high



a offset



a sustain



a end



b vc



b in



b out



b vc x



b start



b high



b offset



b sustain



b end



⠠⠠⠠⠠⠠⠠⠠⠠ NEG

⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ negative slew



⠠⠠⠠⠠⠠⠠⠠⠠ a vc



⠠⠠⠠⠠⠠⠠⠠⠠ a in



⠠⠠⠠⠠⠠⠠⠠⠠ a out



⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ a vc x



⠠⠠⠠⠠⠠⠠⠠⠠ a end



⠠⠠⠠⠠⠠⠠⠠⠠ a out



⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ a offset



⠠⠠⠠⠠⠠⠠⠠⠠ a end



⠠⠠⠠⠠⠠⠠⠠⠠ b vc



⠠⠠⠠⠠⠠⠠⠠⠠ b in



⠠⠠⠠⠠⠠⠠⠠⠠ b out



⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ b vc x



⠠⠠⠠⠠⠠⠠⠠⠠ b end



⠠⠠⠠⠠⠠⠠⠠⠠ b out



⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ b offset



⠠⠠⠠⠠⠠⠠⠠⠠ b end

ENV

envelope generator



end



out



window



start in



out



window offset



cycle



hold



window vc



duration vc



rise



fall



duration x

# ADP adapter



a out



a in



b out



b in



c out



c in

# NOI

noise source



white



white



pink



pink



s/h ac



s/h dc



stepped ac



stepped dc



manual



trigger

# VCF filter


  
 high


  
 band


  
 low


  
 Q



  
 freq vc

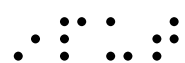


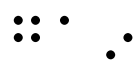
  
 freq vc ×

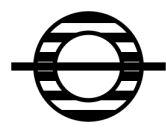
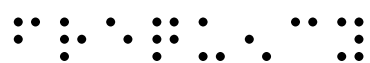

  
 high


  
 band


  
 low


  
 input


  
 gain


  
 frequency

# SSG (1/2)

SSG (1/2)

smooth & stepped generator



smooth vc



smooth out



smooth vc x



smooth cycle



smooth vc offset



smooth hold



stepped vc



stepped sample



stepped vc x



stepped cycle



stepped vc offset



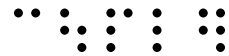
stepped out

## SSG (2/2)

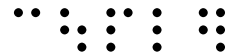
smooth & stepped generator



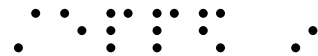
smooth in



coupler



coupler



stepped in

⋮ ⋮ ⋮ ⋮ **SEQ**

⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ sequencer



5 out



5 out



6 out



6 out



4 out



4 out



7 out



7 out



3 out



3 out



8 out



8 out



2 out



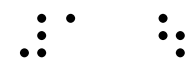
2 out



9 out



9 out



1 out



1 out



10 out



10 out



clock



hold

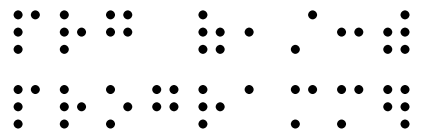


reset



step





# PRG (1/3) programmer



1 out



2 out



3 out



4 out



1 a



2 a



3 a



4 a



1 b



2 b



3 b



4 b



1 c



2 c



3 c



4 c



1 in



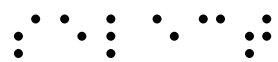
2 in



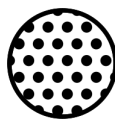
3 in



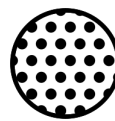
4 in



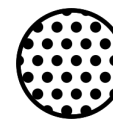
select



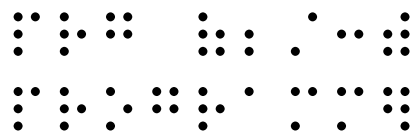
select



select



select



## PRG (2/3) programmer



5 out



6 out



7 out



8 out



5 a



6 a



7 a



8 a



5 b



6 b



7 b



8 b



5 c



6 c



7 c



8 c



5 in



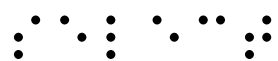
6 in



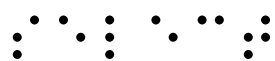
7 in



8 in



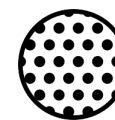
select



select



select



select

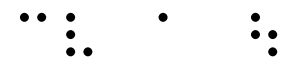
PRG (3/3)  
programmer



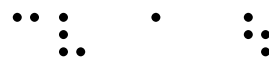
all out



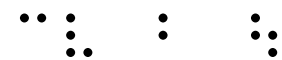
all out



cv a out



cv a out



cv b out



cv b out



cv c out



cv c out

bi-directional router

